

1.4 Classwork: Measurement quiz review

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$

$$1 \text{ in.} = 2.54 \text{ cm} \quad 1 \text{ m} \approx 3.28 \text{ ft} \quad 1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ m} = 100 \text{ cm} \quad 1 \text{ cm} = 10 \text{ mm} \quad 1 \text{ h} = 3600 \text{ s}$$

1. Write down the number of significant digits in each measurement.

(a) 7.25 m 3

(c) 0.045 kg 2

(e) 20.10 cm 4

(b) 3.0 s 2

(d) 105 g 3

(f) $6.0 \times 10^3 \text{ m}$ 2

2. Round each value to three significant figures. Copy the calculator display followed by three dots, then round: $\sqrt{2} = 1.4142135 \dots \approx 1.41$

$$(a) \sqrt{10} = 3.162277\dots \\ \approx 3.16$$

$$(b) \frac{1}{7} = 0.14285714\dots \\ \approx 0.143$$

3. Write in scientific notation, rounded to three significant figures.

(a) The distance from the Earth to the Sun, 149,600,000 km

$$\approx 1.50 \times 10^8 \text{ km}$$

(b) The thickness of a sheet of paper, 0.000102 m

$$\approx 1.02 \times 10^{-4} \text{ m}$$

4. Write each value as an ordinary decimal number.

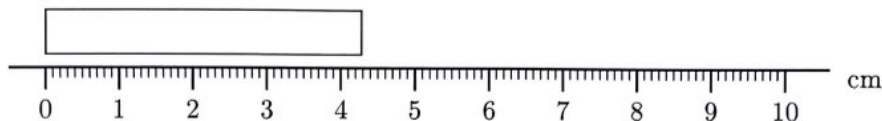
(a) The height of Mount Everest, $8.849 \times 10^3 \text{ m}$

$$8849 \text{ m}$$

(b) The mass of a paper clip, $1.0 \times 10^{-3} \text{ kg}$

$$0.0010 \text{ kg}$$

5. The bar below is measured with a centimeter ruler marked in millimeters.



- (a) Record the length of the bar. Include one estimated digit — the last digit, the one you are not certain of.

4.28 or 4.29 cm

- (b) Write the same length in millimeters, and again in meters.

42.8 mm 0.0428 m

6. Convert between units, rounding to three significant figures. Write the conversion as a fraction and show the units cancelling.

- (a) The Leaning Tower of Pisa is 56.7 meters tall. What is its height in feet?

$$56.7 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 185.976... \text{ ft} \approx 186 \text{ ft}$$

- (b) A sheet of US letter paper is 11.0 inches long. What is its length in centimeters?

$$11.0 \text{ in} \times \frac{2.54 \text{ cm}}{1 \text{ in}} = 27.94 \text{ cm} \approx 27.9 \text{ cm}$$

7. A student walks 60.0 meters down the hallway in 45.0 seconds. Find the average speed in meters per second, rounded to three significant figures.

$$v = \frac{d}{t} = \frac{60.0 \text{ m}}{45.0 \text{ s}} = 1.333... \text{ m/s} \approx 1.33 \text{ m/s}$$

8. A Category 1 hurricane has winds of at least 119 kilometers per hour. What is this speed in meters per second? Write the conversion as a single chain of fractions.

$$119 \frac{\text{km}}{\text{hr}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ s}} = 33.055... \text{ m/s} \approx 33.1 \text{ m/s}$$